Projects:

Project 1: Car Price Prediction System

- Developed a supervised machine learning model to predict car prices using historical sales data.
- Performed data cleaning, feature engineering, model training, and evaluation to ensure high predictive accuracy.
- Deployed the model via Flask for real-time price estimation.
  Technologies: Python, Pandas, Scikit-learn, Flask

Project 2: Retail Sales Forecasting

- Built a time-series forecasting system to predict monthly sales trends for an e-commerce platform.
- Implemented ARIMA and machine learning-based models to capture seasonal patterns and improve prediction accuracy.
  Technologies: Python, Pandas, Statsmodels, Scikit-learn

Project 3: Retrieval-Augmented Generation (RAG) Chatbot

- Designed a document-based question answering system leveraging LangChain, vector embeddings, and large language models.
- Enabled retrieval of relevant document chunks before generating accurate responses, improving user query resolution.
  Technologies: Python, LangChain, FAISS, ChromaDB, OpenAI API

Project 4: Student Performance Prediction

- Developed a predictive analytics system to identify students at risk of poor academic performance using demographic and academic data.
- Applied classification models to support early intervention strategies.
  Technologies: Python, Scikit-learn, Pandas

Project 5: Forex Trading Strategy Automation

- Built an automated system for generating and executing Forex trading strategies based on predictive models.
- Integrated machine learning predictions with trading platforms for real-time execution and performance optimization.
  Technologies: Python, Pandas, Scikit-learn, MetaTrader API

Project 6: Hybrid Car Scanner Automation

- Developed an automated scanning system for hybrid vehicles to analyze diagnostics, performance metrics, and component health.

- Integrated hardware and software components to enable efficient and accurate vehicle assessment.
  Technologies: Python, OBD-II Protocol, Pandas, Flask